Nirmal Prajapati

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology, Kanpur	6.24/10
2021	Gujarat State Board(GSEB)(XII)	Parth School Of Science And Competition, Vadodara	90.8%
2019	Gujarat State Board(GSEB)(X)	Zen School, Padra	93.83%

Scholastic Achievements

- Secured **All India Rank 1049** in **JEE Advanced 2021** among 1,50,000+ shortlisted candidates
- Secured **All India Rank 2644** in **JEE Mains 2021** among 10,00,000+ candidates
- Secured **All India Rank 248** in **Gujarat Common Entrance Test 2021** among 1,00,000+ candidates
- Among **top 1000** students who appeared for **Gujarat State Education Board class(XII) 2021** examination
- Received certificate for Innovation in Science Pursuit for Inspired Research scholarship **(INSPIRE)** under **GSEB**

Key Projects

- **Federated Learning Based Content Caching in Edge Computing** | *Prof Rajesh Hegde, IITK* (*May'23-Ongoing*)
  - Employed **federated learning** framework using **TensorFlow federated** to develop a privacy-preserving content caching scheme which allows users to train locally on their data and share model parameters with the server
  - Implemented caching algorithms like **random, most frequently used, m-ϵ greedy** and **thompson sampling**
  - **Implemented a stacked Autoencoder** approach, which involves training a **neural network** with a single hidden layer to predict the movies present in dataset, which are popular among users based on the learned representations
  - **Concluded** through experimental results that our algorithm **outperforms** the existing caching algorithms
- **MapGo Web Application** | *Course project under Prof. Indranil Saha, CSE Dept.* (*Jan'23-Apr'23*)
  - Developed a web application using the **MERN stack**, showing **campus events** and providing a **personalized map**
  - Collaborated with a team of 10 to design and document everything in the Software Requirements, Specification Document, Design Document, Implementation Document, Testing Document and User Manual for our software
  - Integrated **MongoDB** as the backend for storage of various data. Used the **Mapbox API** for live map functionality
  - Implemented user schemas to efficiently manage **two distinct sub-profiles**, namely **Host and User**, enabling hosts to effortlessly **update event details** while providing users with the ability to **subscribe to their preferred hosts**
- **Bubble Machine** | *Course project under Prof. Urbi Chatterji, CSE Dept.* (*Jan'23-Apr'23*)
  - Created a **Verilog** processor with a custom Instruction Set Architecture, featuring a 32-bit word and instruction size
  - Designed a **FSM** for control signals to guide the processor to Fetch, Decod, Execute instructions using the **ALU**
  - Implemented **Bubble Sort** MIPS code and tested the processor using various inputs, and storing the resulting output
- **Application of Reinforcement Learning** | *Association of Computing Activities, IIT Kanpur* (*May'22-Jul'22*)
  - Learnt **basics of python and supervised machine learning**, **Implemented basics of reinforcement learning** through assignments and excercises, and got familiar with various libraries like **Numpy, Pandas** & Matplotlib
  - Learnt prediction processes like, **Marchovnicov decision process**, **model free prediction** and **model free control**

Technical Skills

Programming Languages	Libraries	Softwares and Tools
C, C++, Python, Verilog HDL	Numpy, Pandas, Matplotlib, TensorFlow	Jupyter Notebooks, Git, LaTeX

Relevant Courses

Data Structure and Algorithms	Fundamentals of computing	Discrete Mathematics
Probability for Computer Science	Introduction to Electronics	Logic in Computer Science
Software developement and operations	Linear Algebra and differential equations	Computer Organization

Positions of Responsibility

- **Senior Executive, Events-Techkriti'23** (*Oct'22-Mar'23*)
  - Helped in the **registration** of more than **1000 participants** in Techkriti **Robogames competitions**, who were interested in various dimensions of Robotics, and also helped in **smooth conduction of Manoeuvre competition**
  - Coordinated with the whole team of **Techkriti'23** containing **200+ students** for smooth conduction of events
- **Senior Executive, Hindi Literary Events-Antaragni'22** (*Oct'22-Mar'23*)
  - Invited **20+ renowned personalities as judges**, Helped HLE team in conduction of total **5 events in 3 days**
  - Coordinated intricate **logistical details of all 5 events**, such as availability, capacity, and accessibility

Extra-Curricular Activities

- Participated in High-jump event in **Udgosh'22**, the anual sports festival of IIT Kanpur
- Part of the Runner up Athletics Team in **Inferno'23**, an inter-hall sports event